

An Immersive Technical Document - Applying Game Design Techniques to Immersive Visualization Laboratories

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ABSTRACT

Immersive visualization laboratories (IVL) offer a wide range of immersive hardware, software and expertise, e.g. virtual reality (VR) to extended reality (XR). Each IVL offers a unique hardware and software infrastructure depending on their focus and use-cases. Not only are these laboratories highly specialized, so is the development process for XR applications. One common use case for IVLs is the collaboration with a domain scientists, each with their own data formats and requirements with regard to output hardware. Based on development processes often employed in the gaming industry, we examine the concept of game design documents (GDD) and developed an immersive technical document (ITD) to optimize immersive application development process. The ITD can mitigate scope creep and help communication between IVL experts and external partners. We have used our ITD in practical courses and general projects and examine advantages and challenges of the document.

Index Terms: Virtual Reality, Large-Scale Immersive Installation, Immersion, Development, Game Design, Development

1 INTRODUCTION

Immersive visualization laboratories (IVL) offer a wide range of immersive hardware, software and expertise [12, 15, 8, 19]. Development in an IVL is therefore as unique as the hardware and software offered, ranging from Virtual Reality (VR) to Extended Reality (XR). There is research on challenges when developing immersive applications [1] as well as workflows [25, 16]. The challenges within development include the difficulty of knowing where to start, difficulty to design for physical aspect of immersive experiences (how natural the immersive application feels) in addition to user testing and evaluation [1]. While IVL experts have experience in developing immersive applications, there are tools to assist in the development of immersive applications.

We examined development processes within the gaming industry, specifically the use of Game Design Documents (GDD) as a starting point for development [18]. GDDs are created at the start of the development and act as a road map to guide development of an application. They help define the scope of the application, establish a clear goal for the application thus reduce rework and disagreements between teams [17]. We find many parallels with the process of developing immersive applications within an IVL and game development.

There is little research on GDDs especially in the context of VR. A search through the IEEE Xplore database produced 18 results for "game design document" but little on GDDs and immersive media.

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Based on GDDs and our experience with software development within an IVL, we propose an immersive technical document (ITD) to optimize the development process of immersive applications. The document is divided into seven main chapters: Overview, Assets, Mechanics, Dynamics, Aesthetics, Technical Details and Misc. This paper will present these chapters as well as the application of the ITD. The goal of the ITD is to be used as a guideline to discuss all aspects in the development of applications within an IVL with all parties involved. This includes mitigating rework, managing expectations and grant transparency between all parties. After the presentation of the ITD, the discussion highlights the advantages and challenges of using the ITD based on various use-cases.

2 BACKGROUND

Many scientific fields can benefit from immersive visualizations (IVs) [10], however, there are many solutions when it comes to creating an IV [11], e.g. software specifically for scientific visualization (SciVis) [4] or game engines [12].

Game design includes all aspects required for creating a game, e.g. goal of the game, player details, story, game play, interfaces and visuals [18]. There are many tools that can assist in the game design process, such as GDDs. This document helps communicate the vision for a game in sufficient detail to implement it [20, 14]. Next to normal games, GDDs have been applied to pedagogical games [22, 24], games to preserve cultural heritage [28] and serious games [5].

However, there is no standardization of GDDs, with various games calling for various forms of GDDs [3]. GDDs require a certain amount of effort to create and maintain, which is why their usefulness has also been discussed in depth [3]. Variations of the GDD include the following: The software requirement specifications (SRS) has been combined with GDD, so expand the GDD to specify more technical aspects or engineering requirements of projects [23]. There are also combinations of GDDs, software engineering methodologies and instructional design [2]. Short Game Design Documents as well as a short GDD for education have been proposed to mitigate effort put into maintaining the document as well as adding domain specific (educational aspects), by simplifying and shortening the length of the document [14]. GDDs specific to educational games has also been proposed (Ped-GDD) [21, 22]. Some research expand the document of a GDD, typically filled out with text, through canvases to encourage creative input [13]. There is also research that examines the use of GDDs for multi-player VR development and added the need to change aspects of GDDs for the multiplayer and VR features. These include immersion and comfort among other aspects [9]. In general GDDs are templates or structured documents that can be filled in with text or other input [21, 23]. As there is no standardization the structure can vary between each GDD.

Advantages of GDDs include reducing rework during development, elimination of hype, clarity and certainty and ease of drafting schedules [20, 17]. Disadvantages include the time consumption it takes to establish and maintain the document [26].

Next to GDDs there is the MDA framework [7]. This framework divides the game design process into three main components: Me-

chanics, Dynamics and Aesthetics (MDA) [7]. Prior research has examined combining the MDA framework to GDDs [17, 26, 23]. The goal of the MDA framework is to bridge the gap between game designers/developers and players, to communicate the vision of the developers to the player [7]. Mechanics describe data representation, dynamics describe run-time behavior of mechanics depending on the player and aesthetics describe emotional responses evoked in the player [7].

3 THE IMMERSIVE TECHNICAL DOCUMENT

There is little research on the application of GDD for an IVL even though it has found application in general software development [23], educational application development [21] and scientific visualization [27]. We found 18 papers on GDDs in the IEEE Xplorer library but little focus on immersive applications.

We created the ITD as a template for the development of immersive applications for IVLs based on GDDs and the MDA framework [7, 23]. The template is generic to suit general XR development, collaborations with domain scientists as well as other immersive media. Our goal was to create a short descriptive document anyone can use for their development.

For this we have structured the ITD into the following chapters: OVERVIEW, ASSETS, MECHANICS, DYNAMICS, AESTHETICS, TECHNICAL DETAILS and MISC (see Table 1). Currently, the ITD is available as a simple structure (see Appendices A) that can be filled in with text for most chapters and flowcharts, mood boards and sketches can be added additionally.

1. **OVERVIEW:** This chapter outlines the general information of the project from title, abstract (1-2 sentences describing the application), involved parties, terminology, immersive medium (AR, MR and/or VR) as well as project deadline. This chapter is to be filled in with text. The goal is to summarize the most important information about the application in a way that can be used for project pitches and as a road map for development. The immersive medium differs between IVLs and should be adapted accordingly. We advise to also discuss target audience in this chapter. The target audience can be general public, domain scientists, etc. and based on the target audience some interactions may need to be adapted.

2. **ASSETS:** This chapter addresses the assets or components within the application and can be filled in with text. There are three main sections: *data*, *user interface (UI)*, *audio* and *other*. *Data* is composed of two subsections we found most relevant within the IVL: Research data and general 3D data. Research data is for domain specific data and requires a documentation of file format, size, location, specifications for that data type and pre-processing if required. This research data can include raw-data but also pre-processed data depending on the project. General 3D data includes 3D-models built additionally for the research data or from a library. 3D-models include but are not limited to interactive objects, non-interactive objects as well as avatars. *UI* includes a list of any user interface required for the application, e.g. main menu, additional menus or interfaces. This sections offers an overview of those UI. *Audio* includes a list of any audio required in the application, if at all, e.g. sound effects or music. *Other* includes assets that do not fall into one of the other categories.

3. **MECHANICS:** This section describes the user's journey. This chapter can be filled in through text or described in a flowchart (see Figure 1) to highlight the users journey through the application. Depending on the goal of the application the user's journey could have a beginning with the option of having an end or only describe the user tasks within the application.

4. **DYNAMICS:** This chapter should be filled out with text. Within this chapter the interactions as well as restrictions within the application are defined. The interactions are split into the sections navigation and selection/manipulation (see Table 1). Selection and manipulation are relevant for both world/objects and any

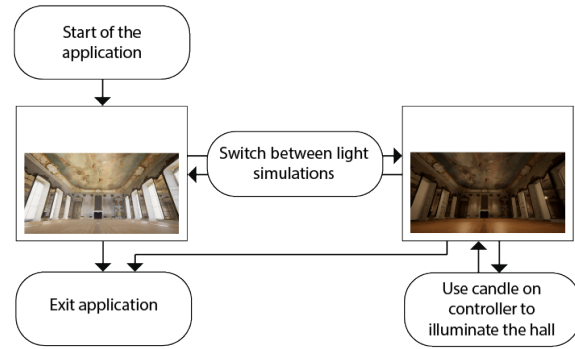


Figure 1: Example: Within the Mechanics Chapter a user's journey of an interactive immersive visualization of a mirror hall [6]

graphical user interface (GUI). Here, the exact interaction needs to be described for clarity between all parties of the project (developers and users). Further, we ask for any restrictions or constraints, e.g. boundaries or number of selected items, within the application.

5. **AESTHETICS:** As VR is a very visual medium, the AESTHETICS chapter in the ITD contains a description of the overall visuals of the application. Here, visuals can be described, e.g. photorealistic or a more cartoon style for educational applications involving students. The description of the overall visuals of the virtual environment (VE) as well as the GUI should be added here. There is a condition for hardware restrictions to be noted, that could influence the visual style. Finally, depending on the application itself there is a section to describe or sketch out the VE to gain a better understanding of the application. This chapter can be filled in with text but also through mind maps and sketches.

6. **TECHNICAL DETAILS:** Here, we document the technical details of the application. Target hardware as well as development software with major version numbers should be noted. This chapter should be filled in through text.

7. **MISC:** The Miscellaneous chapter is for additional thoughts and ideas for future development.

3.1 Application of the ITD

The ITD should be filled out at the start of a project with all stakeholders involved. This includes domain scientists, IVL experts and anyone further involved in this project (developers, etc.) and should be part of an initial meeting.

The initial meeting should be used to fill in the document together and the session should be moderated. For our ITD, these meetings have ranged from 30 minutes to 2 hours. Within the OVERVIEW (see Table 1) it is essential to define a common vision for the project, manage expectations and discuss common terminology between the domain scientists and visualization experts. Following this the chapters ASSETS, MECHANICS, DYNAMICS as well as TECHNICAL DETAILS are essential to discuss with everyone in the initial meeting, this can assist in transparency and communication throughout the teams. The AESTHETICS chapter is very subjective to the project and depends on what domain scientists require, e.g. photo-realism. If there are any hardware restrictions that could effect the visuals the ITD can be used to communicate these restrictions and brainstorm the visuals for the project. Finally, MISC offers space for further thoughts and ideas that may not end up within the scope of the project.

For developers this document can be used as an orientation, documentation and guide throughout the development and for domain scientists it can be used for project pitches, especially the OVERVIEW Chapter.

Chapter	Sections	Description	Format
1. Overview	1.1. Title 1.2. Abstract 1.3. Project Partners 1.4. Terminology 1.5. Immersive Medium 1.6. Deadline	Overview of application	Text
2. Assets	2.1. Data: Research Data and 3D data 2.2. User Interface 2.3. Audio 2.4. Other	Documentation of assets used within the project with sources	Text
3. Mechanics		Description of the user's journey within the application	Text, flowchart or sketch
4. Dynamics	4.1. Navigation 4.2. Selection/Manipulation: World/objects and GUI 4.3. Restrictions	Description of various dynamics within the application	Text
5. Aesthetics	5.1. General Aesthetics: Environment and GUI 5.2. Hardware Restrictions 5.3. Layout of the VE	Description and summary of aesthetics	Text, flowchart, sketch or mind map
6. Technical Details	6.1. Target Hardware 6.2. Software	Description and documentation of technical details	Text
7. Misc		Space for further ideas or notes	Text, flowchart or sketch

Table 1: Structure of the ITD: Chapters and sections with a short description and format

If major issues should arise throughout the development process, e.g. a software that was initially part of the project (TECHNICAL DETAILS) is no longer available, changes should be made to the ITD with a solution and only with the involvement of all parties.

3.2 Use Cases

We have applied our ITD at our IVL, within projects with domain scientists as well as teaching, where student developed projects for CAVE and VR HMDs.

Our practical courses involve group or individual projects of master students within the span of one university semester. Due to the short development time frame, we have discovered that using the ITD at the beginning of the semester assists groups in defining their goal. The students discuss in groups or brainstorm individually (depending on if it was for a group project or an individual project) and document their ideas in the ITD. They then present their ideas to the IVL experts and discuss feasibility of the project in the given time frame.

For the projects with domain scientists, these discussions are also used to clarify the vision and fuel communication between parties. The discussions aid in creating transparency of the development process within the IVL to domain scientists, highlighting the effort and work that is required.

4 DISCUSSION

Throughout these collaborations we have identified certain challenges and advantages.

4.1 Advantages

One major advantage of using the ITD is having all parties involved discuss the vision of the application. These discussions and the ITD have aided all parties in mitigating scope creep. A phenomenon, where the scope of a project can exceed the budget (monetary and time) during the development process. As no changes should be

made to the ITD without discussion within the team, no sudden changes should be made. This aligns with advantages of GDDs or reducing rework [17].

We have discovered the management of expectations with our projects with domain scientists have reduced frustration during development or at the end result.

The OVERVIEW chapter of the ITD can be used as a project pitch page for all stakeholders, similar to a GDD.

In terms of documentation, the ITD documents major aspects of the application, which in turn assists developers in the development phase, e.g. data and interactions with in DYNAMICS.

4.2 Challenges

The ITD is not without challenges. While we had initial meetings with domain scientists before the ITD, we have discovered that with the document the initial meeting requires effort for everyone involved and it needs to be moderated.

Another major issue with the ITD is similar to the issue with classic GDDs or general VR projects: maintenance. On the one hand the maintenance of the document should be held upright by one person (not everyone editing the same document) and edits should be made as a group. On the other hand the ITD does not reflect or consider the maintenance of the project itself. Here, we are looking into a new chapter in the document for major updates (be it software or data).

4.3 Limitations

Next to the challenges, the ITD has some limitations. First, it is based on our experience developing immersive applications within an IVL and we are not aware of any other IVL using a ITD or similar document.

5 CONCLUSION AND FUTURE WORK

We propose an immersive ITD to aid in the development of applications for IVLs based on prior work with a focus on GDD. We describe the document itself and how to apply it as well as discuss advantages and challenges we identified.

We currently have the immersive ITD as a TXT file with the structure proposed in this paper (see Table 1 and Appendix A for a version that can be used) but in future we want to develop a web form to simplify the establishment of the document. We have made the ITD available and would like to gather data on if the ITD could assist other IVLs in their development processes.

We also want to perform expert interviews on the document itself and compare development with and without the document.

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A APPENDIX A

Basic structure of the ITD to be used.

1. Overview
 - 1.1 Title
 - 1.2 Abstract (1-2 Sentences)
 - 1.3 Project Partners
 - 1.4 Terminology
 - 1.5 Immersive Medium
 - 1.6 Deadline
2. Assets
 - 2.1 Data: Research Data and 3D Data
 - 2.2 User Interface
 - 2.3 Audio
 - 2.4 Other
3. Mechanics
4. Dynamics
 - 4.1 Navigation
 - 4.2 Selection/Manipulation: World/objects and GUI
 - 4.3 Restrictions
5. Aesthetics
 - 5.1 General Aesthetics: Environment and GUI
 - 5.2 Hardware Restrictions
 - 5.3 Layout of the Virtual Environment
6. Technical Details
 - 6.1 Target Hardware
 - 6.2 Software
7. Misc

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